

Operating Systems

Lab Assignment 4

Shell Scripting -Array, Loop

<u>Roll. No.</u>	<u>Gr. No.</u>	<u>Div</u>	<u>Batch</u>	<u>Name</u>
21	1201079 6	D	1	Shubham Swapneel Shah

Problems to be solved in the lab:

1. Write Shell script to find out positive and negative numbers from accepted array. Assume Array consists of 5 numbers. Also accept array from user.

```
-2 is an Negative number
shubham@Shubham-Shah:~/Lab3$ nano Lab3_1.sh
shubham@Shubham-Shah:~/Lab3$ bash Lab3_1.sh
Enter Array Size
5
Enter Array elements
-1
2
0
-3
-4
Entered Array elements =
-1 2 0 -3 -4
Positive numbers
2 is an Positive number
0 is an Positive number
Negative numbers
-1 is an Negative number
-3 is an Negative number
-4 is an Negative number
```

```
#bin/bash

#Accepting size
echo "Enter Array Size"
read size

#Accepting array elements
echo "Enter Array elements"
for((i=0;i<$size;i++))
do
read b
c[$i]=$b
done

#Displaying Array
echo "Entered Array elements = "
echo ${c[@]}

num=0
#Displaying Positive and Negative Numbers
echo "Positive numbers"
for((i=0;i<$size;i++))
do
t=${c[$i]}
if [ $t -ge $num ]
then
echo "$t is an Positive number"
fi
done

echo "Negative numbers"
for((i=0;i<$size;i++))
do
t=${c[$i]}
if [ $t -lt $num ]
then
echo "$t is an Negative number"
fi
done
```

2. Write Shell script to find out even and odd numbers from accepted array. Assume Array consists of 5 numbers. Also accept arrays from users.

```
shubham@Shubham-Shah:~/Lab3$ nano Lab3_2.sh
shubham@Shubham-Shah:~/Lab3$ bash Lab3_2.sh
Enter Array elements
1
2
3
4
5
Odd nos are
1
3
5
Even nos are
2
4
shubham@Shubham-Shah:~/Lab3$
```

```
#bin/bash

#Accepting array elements
echo "Enter Array elements"
for((i=0;i<5;i++))
do
read b
c[$i]=$b
done
num=0
#Displaying Even and Odd numbers
echo "Odd nos are"
for((i=0;i<5;i++))
do
t=${c[$i]}
mod=$((t%2))
if [ $mod -gt $num ]
then
echo "$t"
fi
done
echo "Even nos are"
for((i=0;i<5;i++))
do
t=${c[$i]}
mod=$((t%2))
if [ $mod -le $num ]
then
echo "$t"
fi
done
```

3. Write Shell script to sort array numbers ascending and descending order. Assume Array consists of 5 numbers. Also accept arrays from users.

```

10 7 14 1 11
shubham@Shubham-Shah:~/Lab3$ nano Lab3_4.sh
shubham@Shubham-Shah:~/Lab3$ bash Lab3_4.sh
Enter the array of 5 elements
11
1
10
14
7
Sorted array in descending order is
14 11 10 7 1
Sorted array in ascending order is
1 7 10 11 14

```

```

#!/bin/bash
echo "Enter the array of 5 elements"
for((i=0;i<5;i++))
do
    read b
    a[$i]=$b
    c[$i]=$b
done
echo "Sorted array in descending order is"
min=${a[0]}
max=${a[4]}
for((i=0;i<5;i++))
do
    for((j=0;j<4;j++))
    do
        if [ ${a[$j]} -lt ${a[$((j+1))]} ]
        then
            temp=${a[$j]}
            a[$j]=${a[$j+1]}
            a[$((j+1))]=$temp
        fi
    done
done
echo ${a[@]}
echo "Sorted array in ascending order is"
min=${c[0]}
max=${c[4]}
for((i=0;i<5;i++))
do
    for((j=0;j<4;j++))
    do
        if [ ${c[$j]} -gt ${c[$((j+1))]} ]
        then
            temp=${c[$j]}
            c[$j]=${c[$j+1]}
            c[$((j+1))]=$temp
        fi
    done
done
echo ${c[@]}

```

4. Write Shell script to find out smallest number and largest number of given array. Assume

Array consists of 5 numbers. Also accept arrays from users.

```
shubham@Shubham-Shah:~/Lab3$ nano Lab3_3.sh
shubham@Shubham-Shah:~/Lab3$ bash Lab3_3.sh
Enter Array Size
5
Enter Array elements
2
6
9
0
1
Entered Array elements =
2 6 9 0 1
Max = 9
Min = 0
shubham@Shubham-Shah:~/Lab3$
```

```
GNU nano 4.8
#bin/bash

#Accepting size
echo "Enter Array Size"
read size

#Accepting array elements
echo "Enter Array elements"
for((i=0;i<$size;i++))
do
read b
c[$i]=$b
done

#Displaying Array
echo "Entered Array elements = "
echo ${c[@]}

num=0
max=${c[0]}
min=${c[0]}
#Displaying Positive and Negative Numbers
for((i=1;i<$size;i++))
do
t=${c[$i]}
if [ $t -gt $max ]
then
max=$t
fi
if [ $t -le $min ]
then
min=$t
fi
done
echo "Max = $max"
echo "Min = $min"
```

5. Write shell script to find out the reverse number of a given number.

```
shubham@Shubham-Shah:~/Lab3$ nano Lab3_5.sh
shubham@Shubham-Shah:~/Lab3$ bash Lab3_5.sh
Enter the number to be reversed
1876
Reverse of the number is 6781
shubham@Shubham-Shah:~/Lab3$
```

```
GNU nano 4.8 La
#bin/bash
echo "Enter the number to be reversed"
read num
while [ $num -gt 0 ]
do
reverse=$((reverse*10+$num%10))
num=$((num/10))
done
echo "Reverse of the number is $reverse"
```

6. Write a shell script to create a fibonacci series.

```
0 1 1 2 3 5 8 13
shubham@Shubham-Shah:~/Lab3$ nano Lab3_5.sh
shubham@Shubham-Shah:~/Lab3$ bash Lab3_5.sh
Enter the count of Fibonacci Members
8
Fibonacci elements =
0 1 1 2 3 5 8 13
shubham@Shubham-Shah:~/Lab3$
```

```
#!/bin/bash

#Accepting size
echo "Enter the count of Fibonacci Members"
read size

#Accepting elements
#echo "Enter Fibonacci elements"
a=0
b=1
arr[0]=$a
arr[1]=$b

#Storing the Fibonacci series elements in an array
for((j=2;j<$size;j++))
do
c=$((a+b))
arr[$j]=$c
a=$b
b=$c
done
echo "Fibonacci_elements = "
echo ${arr[@]}
```